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Sliding Mode Control in Aerospace Applications: A Survey

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ABSTRACT

Sliding mode control (SMC) enjoys robustness to matched and unmatched (in the case of minimum phase input-output dynamics) bounded perturbations, and finite time convergence. Second-order and higher-order sliding mode control systems (2-SMC/HOSMC) retain all the advantages of sliding mode control, but in addition can be applied to systems of high relative degree while providing control continuity and higher accuracy, output tracking error stabilization. Adaptive and finite-convergence-time SMC/2-SMC/HOSMC provides more advanced features compared to basic sliding mode control techniques. In this paper, sliding mode control laws are considered for the aerospace applications—including control of damaged aircraft, satellite, and formation satellite control, reusable launch vehicle control, and the guidance and control of hypersonic vehicles and missiles. These applications have been reviewed, and the advantages of sliding mode control techniques, including high accuracy, finite time convergence, and insensitivity to bounded perturbations and parametric uncertainties are confirmed via numerous simulations and implementations.

1 | Introduction

From the earliest days of manned flight, it was recognized that feedback control was essential. Lawrence Sperry demonstrated “hands-free” controlled flight in 1914. It was based on an autopilot which went by the moniker “Metal Mike”, which utilized three gyroscopes and pneumatically activated control surfaces [1]. Over the next century, flight control systems evolved from purely mechanical devices to digital fly-by-wire systems implemented on a flight control computer. Aerospace applications have always served as a testing ground for new developments in terms of control theory, and indeed, these disciplines have a somewhat synergistic dependence. Aerospace systems are inherently nonlinear and are required to operate in a wide range of conditions, and are subject to significant exogenous disturbances. They also (usually) constitute safety-critical systems [2]. Different control paradigms have been adopted to address these challenges, and there is a significant literature on the subject. This paper will

focus on how sliding mode control principles have been adopted and tested in this application domain.

Sliding mode control emerged as a special case of the study of Variable Structure Systems (VSS). In these systems, the feedback control law comprised fixed gain structures, which were deployed together with rules to determine which structure would be used at any one time for specified regions of the phase space. In particular, because of the switching rules, the closed loop systems often presented a dynamical system with a discontinuous right-hand side. The use of Variable Structure Systems (or discontinuous control systems) to potentially create simple automatic flight control systems was studied as far back as the 1940s [3]. At that time, such devices could be reliably engineered, but at this early juncture, a formal theory had yet to be created, and the approach lacked formal design methods. As a particular form of VSS, it was discovered that, as a result of the inclusion of switching/relay terms within a feedback control structure, a collapse in the

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dynamical order of the closed-loop behavior could occur. Such a regime was described as a *sliding motion* [4]. Importantly, such a motion was found to be accompanied by the property of insensitivity to a class of uncertainty and disturbances acting in the control channels—referred to as “matched uncertainty” [5]. This robustness property fueled interest in the development of these ideas—which were first widely disseminated (in English) in [6]. This work, emerging from the former USSR, represented the first attempts to create systematic design and analysis approaches. From a design perspective, the establishment of a sliding mode became viewed as one of forcing a so-called sliding variable σ , formed as a function of the state of the system, to zero in finite time, and then maintaining this constraint for all subsequent time. From a geometric perspective, this can be viewed as forcing the states of the system under feedback control to reach and then remain on the surface in the state space associated with the constraint $\sigma(x) = 0$. Practical design paradigms for the creation of suitable sliding surfaces appear in the monograph by Utkin from the early 1980s [7] and the survey paper by DeCarlo et al. [8]. The latter also proposed the idea of an *integral sliding mode* associated with a particular form of time-varying surface. This approach has the advantage of enforcing and maintaining a sliding motion from time $t = 0$, thus obviating the reaching phase, which typically exists prior to the establishment of the sliding motion. A downside to the introduction of a sliding motion in practical engineering systems was the presence of “chattering”—high frequency oscillations arising from motion in the vicinity of $\sigma = 0$, resulting from imperfections in the implementation process, including finite sampling, unmodeled dynamics, and delays. This phenomenon was seen as a major obstacle to implementation in many practical systems. The “discovery” of *second-order sliding modes* which force both $\sigma = \dot{\sigma} = 0$ in finite time, gave further impetus to the development of the field in the 1990’s [9]. These methods helped mitigate the occurrence of the “chattering phenomenon” and made the approach more appealing to many practitioners. Subsequently, *higher-order sliding modes* were explored in which the sliding variable σ and its first $k - 1$ derivatives are driven to zero in finite time [9, 10]. In principle these higher-order approaches permit the discontinuities to be “hidden” in the derivatives of the control signal, and the chattering effects are greatly reduced. The development of high-order sliding mode ideas led to the creation of arbitrary order robust exact differentiators [10] which have found numerous application areas.

The robustness properties of sliding mode controllers make them suitable for applications in which there is significant model uncertainty or the plant itself is required to operate in a wide range of conditions. Unsurprisingly, these ideas have been applied to aerospace problems. All the sliding mode developments described above have been applied to various aerospace platforms—both manned and unmanned. This paper seeks to provide a survey of some of these applications and (implicitly) presents the evolution and increasing sophistication of the sliding mode method, mirroring the developments described above.

The paper is divided into four main parts. Section 2 explores the fault-tolerant control characteristics of SMC—applied primarily to conventional aircraft designs. The section also discusses how the inherent robustness of SMC can be exploited for fault tolerance. Section 3 focuses on satellite and satellite formation control, and argues that SMC is an effective technique

for combating the unknown bounded perturbations that affect these structures. Section 4 considers applications to control of Reusable Launch Vehicles. It has been shown to be an attractive methodology for RLV because of its inherent insensitivity and robustness to unknown plant uncertainties and external disturbances, and this robustness can accommodate the RLV’s different operating regimes. Finally, Section 5 considers Hypersonic Vehicles. These vehicles feature static instabilities, strong nonlinearities, as well as exhibiting significant uncertainties in aerodynamic coefficients because at hypersonic velocities, the vehicle becomes enveloped in a “plasma sheath”. All the effects combined create huge challenges to the control system design.

2 | Damaged Aircraft Sliding Mode Control

Conventional SMCs for aircraft flight control were studied in [11–16]. Specifically, SMCs for modern military jet-fighters (F-14s and F-18s), were presented in [11] and [12]. In these works, robust flight control laws, based on variable structure systems and sliding mode control (VSS/SMC) were designed for simulations with perturbations of known bounds, based on simplified multi-input-multi-output (MIMO) models of the F-14 and F-18 aircraft. The robustness and finite time convergence of the sliding variables to zero, followed by asymptotic convergence of the commanded output (angle of attack, pitch rate, and sideslip angle) tracking errors to zero in the sliding mode, were guaranteed. An output tracking SMC was designed and studied for a tailless rigid-body aircraft in [13], where pitch, roll rate, and sideslip angle represented the controlled outputs. To achieve continuous control (in terms of the aerodynamic surface deflections), the *sign* functions were replaced by saturation elements. The smoothed SMCs ensured the sliding variable convergences to a boundary layer about the sliding surface, resulting in uniform ultimate boundedness of the tracking errors. Therefore, the tracking accuracy was partially compromised. An SMC was proposed for controlling a damaged aircraft in [14]. In the outer loop, the corrected coordinated turn inversion transformation is accomplished, and then in the inner loops, the angular rate tracking profiles are followed by local decoupled SMCs. An automatic recovery algorithm updates the prescribed maneuver in case of excessive damage to guarantee the sliding mode existence condition, exploiting all the control authority that is left. The efficacy of the proposed SMC flight control system has been verified via simulation. A flat spin recovery strategy for aircraft is proposed in [15] using Sliding Mode thrust vector control. The spin recovery was verified for an F-18 aircraft via simulations. In [16], an integral SMC is used for robust tracking control of the pitch angle command of a damaged flexible aircraft.

A *Reconfigurable SMC* design for aircraft flight control was proposed and studied in [17, 18]. Reconfiguration of the sliding surface boundary layer is used in [17] to account for the deflection and rate limits, and actuator dynamics, and a *reconfigurable* SMC is proposed for achieving robust, high accuracy tracking of outer loop command angles before and after damage. This algorithm has also been applied for *reconfigurable* flight control design in [18] for a tailless jet fighter with redundant control surfaces. Command profiles for angles and angular rate are robustly followed in the inner loop via finite-reaching-time continuous SMCs. Here, continuity of the control signal was achieved by

replacing the sign function by a sigmoid; this slightly sacrifices the tracking accuracy that can be achieved. The work exploits an optimal control allocation algorithm.

SMC for aircraft engines were considered in [19–21]. A constraint-tolerant SMC design was proposed in [19] for improving the performance of an aircraft engine subject to uncertainty. A turbofan engine is used for the simulations, which demonstrate the efficacy of the proposed methodology. Sliding mode control of turbine aircraft engines is also studied in [20] where its applicability is demonstrated via simulations. In [21], a novel SMC-based control scheme for aircraft engines within an acceleration/deceleration limiter is proposed. The simulation results show the efficacy of the proposed SMC in controlling all outputs, including the core shaft acceleration, the high-pressure turbine outlet temperature increment, and the high-pressure compressor stall margin increment.

In the early 2000s, there was considerable interest in the development of Fault-Tolerant-Control strategies by the aerospace commercial sector. The European Union funded a series of industry-led projects largely concerned with fault detection and fault-tolerant control from an analytic redundancy perspective. The overarching objective was to increase the Technology Readiness Levels (TRLs) of “modern” model based Fault Detection and Fault-Tolerant-Control methods. Since actuator faults can often be treated as a special case of matched uncertainty, such problems can be well addressed by SMC methods. SMC combined with a novel form of control allocation was proposed in [22]. In the control allocation framework in [22], control signals are automatically distributed to the healthier members of the actuator suite, based on an assessed level of effectiveness. These ideas were subsequently extended to Linear Parameter Varying systems (LPV) of the form

$$\dot{x} = A(\rho)x + B(\rho)u + D\xi,$$

in which the system and input distribution matrices depend on a scheduling parameter ρ , and the signal ξ represents lumped uncertainties [23]. In [22, 23], it is assumed that the system is overactuated in the sense that there are more control inputs than necessary to create a suitable level of closed-loop performance. The LPV approach to system modeling is widely adopted in the aerospace sector, and the scheduling parameters are usually functions of the mass of the aircraft, its altitude, and the relative center of gravity position (although some controlled variables such as speed and angle of attack, etc., are often included, which technically makes the system Quasi-LPV [24]). In [23], a controller of the form

$$u(t) = W(t)B_2(\rho)^T (B_2(\rho)W(t)^2 B_2(\rho)^T)^{-1} v, \quad (1)$$

is proposed in which $B_2(\rho)$ is a sub-component of $B(\rho)$, and $W(t)$ is a weighting matrix which reflects the “health” of the actuators. Specifically, $W(t)$ is a semi-positive definite diagonal matrix whose diagonal entries capture the health of the corresponding actuators: If w_i is the i th diagonal element of W then it is assumed that $w_i \in [0, 1]$ with the interpretation that if $w_i = 1$ then the i th actuator is fault free and working normally, if $w_i = 0$ then the i th actuator has totally failed, and if $0 < w_i < 1$ then the actuator is subject to some partial fault. To implement the controller in



FIGURE 1 | MuPAL-α is a multi-purpose research aircraft [23].

Equation (1), these quantities must be estimated (see, e.g., [25]). In Equation (1), v represents a virtual control signal (of lower dimension than the “physical” control signal u). The expression in Equation (1) represents an allocator structure that distributes the virtual control signal to the physical actuators. A property of (1) is that if $w_i \rightarrow 0$, then provided $\det(B_2(\rho)W^2 B_2(\rho)^T) \neq 0$ (which is possible since $B_2(\rho)$ has more columns than rows), then $u_i \rightarrow 0$. This means that, automatically, the allocator structure distributes control signal commands away from failing actuators. In Equation (1), the control law is a sliding-mode controller which is designed about the “virtual system” whose inputs are given by v . As part of the GARTEUR AG16 program [2] the controller proposed method in this study was implemented on the SIMONA motion simulator at the Technical University of Delft, which was configured to represent the crippled cargo aircraft at the center of the Bijlmermeer crash [26]. The controller demonstrated that, theoretically, the stricken aircraft could have landed successfully with an understanding of the damage and an appropriate FTC strategy [2, 26]. Hamayun et al. [27] considered the use of Integral Sliding Mode Control (ISMC), combined with Control Allocation, for FTC. In the nominal situation, the ISMC replicates the performance of the original underlying control laws. Only in the presence of actuator faults will the effects of the sliding mode elements be observed as they attempt to retain the original fault-free performance levels. Figure 1 illustrates that MuPAL-α is a multi-purpose research aircraft.

A controller based on this approach was flight-tested on the SIMONA motion simulator [28]. Subsequent projects sought to increase TRLs in terms of the application of robust and adaptive control methods applied to fault-tolerant control problems. Chen et al. [29] describes the results obtained from applying an ISMC scheme to an industrially created benchmark simulation involving the loss of Total Airspace and Angle-of-Attack sensor measurements. The advantage of employing the ISMC (rather than a more traditional SMC) is that it can be retrofitted to an existing in-situ control scheme to engender a higher level of fault tolerance. This is advantageous from the perspective of certification. The last of the triumvirate of EU projects (VISION) in conjunction with NEDO and the Japan Aerospace Exploration Agency (JAXA) saw the *flight testing* of sliding mode fault-tolerant controllers on a piloted aircraft [23, 30].

Second- and higher-order sliding mode control techniques (2-SMC/HOSMC) that can handle systems of relative degree 2 and higher, as well as provide higher stabilization accuracy with respect to classical SMC, represent advanced robust control approaches. With the methods, control chattering is mitigated; however, the discrete-time implementation of 2-SMC and HOSM requires smaller sampling times compared to classical SMC, which can be challenging. *2-SMC and HOSM control* applied to aircraft were investigated in [31–33]. Super twisting control (2-SMC) was employed in [31] for controlling the pitch angle of uncrewed aircraft subject to perturbation bounded derivatives. The Super twisting controller generates a continuous control signal and provides finite time convergence in the presence of smooth perturbations with bounded derivatives. This is advantageous for this application. The computer simulations and flight tests demonstrated high accuracy and robust performance.

Super-twisting (2-SMC) control was also studied in [32] to achieve finite-time stability and appropriate performance for aircraft, based on thrust vectoring, while maneuvering at a high angle of attack. Taking into account a lack of aerodynamic control surfaces, thrust vectoring with double nozzles is added. The disturbances that include coupling between the different channels, aerodynamic uncertainties, and unmodeled dynamics are estimated by a finite-time extended state observer. In [33], aircraft control system certification based on providing the required phase margin and gain margin is extended to higher-order sliding mode (HOSM) controllers. A design algorithm based on a cascaded dynamic compensator for a dynamically perturbed HOSM control system is proposed. In this work, the theoretical developments are validated via simulations controlling the attitude of an F-16 aircraft. The aircraft controller based on the super twisting algorithm enjoys the property of control continuity, finite time convergence, and enhanced stabilization accuracy.

Further enhancement of robust control has been provided by SMC and 2-SMC with adaptation. *Adaptive SMC and 2-SMC* for aircraft were investigated in [34–37]. The controller based on the super-twisting algorithm enjoys control continuity, finite-time convergence, and enhanced stabilization accuracy. A super twisting controller with *adaptation* is employed in [34] for controlling aircraft attitude while using a novel explicit longitudinal model of a wing-sweep morphing aircraft. Adaptive SMC and 2-SMC are very effective for uncertain aircraft systems, whose bounds are not known. The cost of adopting gain adaptation is increased complexity, which is a challenge for implementation. It is designed to account for the uncertainties in the aerodynamics, mass, and inertial properties. Controlling an aircraft landing on an aircraft carrier was accomplished in [35] using an *adaptive* super twisting control taking into account the model uncertainties, deck motion, and air wake disturbances. In [36], an *adaptive* SMC in a backstepping framework is used for attitude control of aircraft subject to perturbations with unknown boundaries. A novel continuous adaptive non-homogeneous supertwisting control for aircraft anti-skid braking system was proposed and studied in [37]. The perturbations were estimated using a fixed-time extended state observer. As a result, the controller gains were not overestimated. Comparative experimental results demonstrate the effectiveness and superiority of the proposed controller over the non-adaptive ones.

Recently proposed fixed- and predefined-time SMC techniques constitute a promising development in robust control. *Fixed and predefined time SMC* is investigated in [38, 39] for aircraft control. A fixed-time anti-saturation adaptive control algorithm is proposed in [38] for addressing the challenge of attitude tracking control in morphing aircraft subject to actuator faults, nonlinearities, and bounded perturbations. The proposed control algorithm consists of a novel nonsingular fixed-time sliding variable, a fixed-time anti-saturation compensator, and a continuous fixed-time adaptive control law. An SMC algorithm with predefined time convergence is proposed and studied in [39] for addressing the attitude control problem in aircraft. A predefined time observer is developed to estimate modeling uncertainties and external disturbances within a predefined time. A predefined time non-singular terminal sliding mode reaching law is then proposed for attitude control of aircraft.

Note that the parameter tuning of fixed and predefined-time controllers for aircraft systems can be complicated. *A MATLAB-based SMC design and simulation software toolbox* that supports aerospace vehicle applications, including aircraft, launch vehicles, UAVs, missile interceptors, and hypersonic missiles, is described and studied in [40]. The SMC Aero libraries include first-order SMC, 2-SMC, and higher-order sliding mode controllers (either fixed gain or adaptive), as well as higher-order sliding mode differentiators. The efficacy of the SMC Aero Toolbox is confirmed in case studies.

Simulation results presented in all the reviewed papers on SMC for damaged aircraft showed the precise attitude tracking of aircraft subject to uncertainties and damage, and bounded external disturbances can be achieved based on SMC, SMC-based SMDO, and HOSMC methods.

3 | Satellite and Satellite Formation Control

Satellite formations (SF) are used in many application fields, including astronomy, geodesy, Earth surveillance, and space exploration [41, 42]. Following [41, 42], satellite formation (SF) flight can be defined as the movement of a group of more than one spacecraft, whose dynamics are controlled by a common control law. Control of SFs is challenging because of model uncertainties and external disturbances (perturbations) that complicate the successful accomplishment of the control task. Sliding mode control [9, 43, 44] represents a powerful technique for SF flight control in the presence of bounded perturbations—with possibly bounded derivatives (see, for instance, [45–59]). Note that these bounds may be unknown. (Of course, de facto, individual satellites affected by perturbations can also be effectively controlled by SMC as well [60–70].)

The design and performance of an SMC for a perturbed SF antenna array with dual quaternion dynamics that describe simultaneously the translational and rotational motion of a rigid body [71], is considered in [45, 48]. A configuration of satellites as a distributed antenna array is presented in Figure 2, where λ is the wavelength of the signal and ξ_i is the separation between satellites i , with angle of incidence θ .

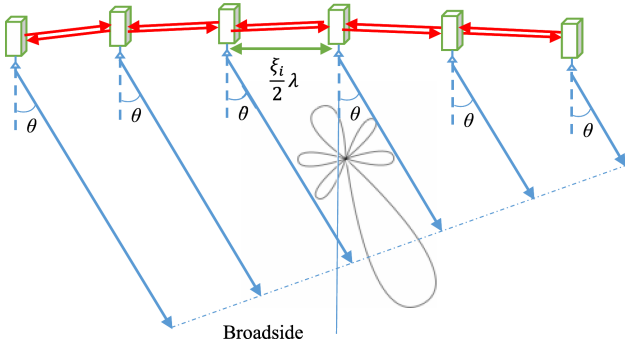


FIGURE 2 | Configuration of satellite formation antenna array [45].

Three adaptive continuous dual quaternion-based SMC algorithms for controlling a perturbed satellite antenna array are investigated in [45, 48]. The perturbations and their derivatives are assumed to be bounded, but the bounds themselves are unknown. The adaptive SMC algorithms are capable of handling coupled perturbed systems, including dual quaternion models. Note that the use of quaternions in the kinematic equations helps avoid singularities in the satellite rotation matrix [72]. The dual quaternion kinematic equations are taken as [72]:

$$\begin{aligned}\dot{\hat{q}}_{BI} &= \frac{1}{2} \hat{\omega}_{BI}^I \hat{q}_{BI} = \frac{1}{2} \hat{q}_{BI} \hat{\omega}_{BI}^B \\ \dot{\hat{q}}_{DI} &= \frac{1}{2} \hat{\omega}_{DI}^I \hat{q}_{DI} = \frac{1}{2} \hat{q}_{DI} \hat{\omega}_{DI}^B,\end{aligned}\quad (2)$$

where $\hat{q}_{YZ}$ is a dual quaternion describing the relative position of the Y frame with respect to the Z frame, $\hat{\omega}_{YZ}^X$ is the dual relative angular velocity of the Y frame with respect to the Z frame defined in the X frame, such that

$$\hat{\omega}_{YZ}^X = \omega_{YZ}^X + \epsilon(v_{YZ}^X + \omega_{YZ}^X \times r_{YZ}^X),$$

and $v_{YZ}^X = (\bar{v}_{YZ}^X, 0)$ where ϵ is the *dual unit* defined as $\epsilon^2 = 0$, and $\epsilon \neq 0$, and $\bar{v}_{YZ}^X$ is the linear velocity of the Y frame with respect to the Z frame defined in the X frame.

The dual error quaternion is given in [45–48, 73] as

$$\hat{q}_{BD} := \hat{q}_{DI}^* \hat{q}_{BI} = q_{BD} + \epsilon \frac{1}{2} q_{BD} r_{BD}^B, \quad (3)$$

and the error kinematics are derived as

$$\dot{\hat{q}}_{BD} = \frac{1}{2} \hat{q}_{BD} \hat{\omega}_{BD}^B = \frac{1}{2} \hat{\omega}_{BD}^D \hat{q}_{BD}, \quad (4)$$

where $r_{BD}^B = r_{BI}^B - r_{DI}^B$, and $\hat{q}_{BD} \in \mathbb{H}^u$, $\hat{\omega}_{BD}^B = \hat{\omega}_{BI}^B - \hat{\omega}_{DI}^B$ is the dual relative angular velocity between the body frame and the desired frame as represented in the body frame. The dual error quaternion is a unit dual quaternion that represents the rotation q_{BD} and translation r_{BD}^B that are required to align the body frame of the chaser satellite to its desired frame.

The combined kinematics/dynamics for each element of the satellite distributed satellite antenna array formation (extendable to N -satellites with appropriate guidance laws and coordination) may be rewritten as

$$\begin{cases} \dot{\hat{q}}_{BD} = \frac{1}{2} \hat{q}_{BD} \hat{\omega}_{BD}^B, \\ \left(\dot{\hat{\omega}}_{BD}^B \right) = (M^B)^{-1} \star \left\{ \hat{f}_c^B + \hat{f}_d^B - (\hat{\omega}_{BD}^B + \hat{\omega}_{DI}^B) \right. \\ \quad \times \left(M^B \star \left((\hat{\omega}_{BD}^B)^s + (\hat{\omega}_{DI}^B)^s \right) \right) \\ \quad \left. - M^B \star \left(\hat{q}_{BD}^* \hat{\omega}_{DI}^D \hat{q}_{BD} \right)^s - M^B \star \left(\hat{\omega}_{DI}^B \times \hat{\omega}_{BD}^B \right)^s \right\} \end{cases}, \quad (5)$$

where M^B is the dual inertia matrix defined in [73] as

$$M^B \triangleq \begin{bmatrix} mI_3 & 0_{3 \times 1} & 0_{3 \times 3} & 0_{3 \times 1} \\ 0_{1 \times 3} & 1 & 0_{1 \times 3} & 0 \\ 0_{3 \times 3} & 0_{3 \times 1} & \bar{I}^B & 0 \\ 0_{1 \times 3} & 0 & 0_{1 \times 3} & 1 \end{bmatrix}$$

and $\star$ is the quaternion operation defined in [73]. In the above, m is the mass of the chaser satellite, I_3 is a 3×3 identity matrix, and $\bar{I}^B \in \mathbb{R}^{3 \times 3}$ is the mass moment of inertia of the body about its center of mass written in the body frame

$$\bar{I}^B = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{bmatrix}.$$

In Equation (5), the term $\hat{f}_c^B = f_c^B + \epsilon \tau_c^B$ represents the dual control forces and $\hat{f}_d^B$ represents the perturbation whose bounds and the bounds on its derivatives exist but are not known, that is,

$$\|(M^B)^{-1}\| \|\hat{f}_d^B\| \leq f_0 \quad \text{and} \quad \|(M^B)^{-1}\| \|\dot{\hat{f}}_d^B\| \leq f_1,$$

where $f_0 > 0$ and $f_1 > 0$ are unknown constants. Also note the superscript s indicates the *swap* of the dual quaternion, that is, given $\hat{\omega} = \omega_r + \epsilon \omega_d$, we have $\hat{\omega}^s = \omega_d + \epsilon \omega_r$.

The control function $\hat{f}_c^B$ is to be defined in terms of an adaptive unit vector SMC, and a second-order SMC (2-SMC) with non-overestimated control gains to drive the error states, $\hat{q}_{BD} \rightarrow \pm \hat{1}$ and $\hat{\omega}_{BD}^B \rightarrow \hat{0}$ (in other words, the error between the chaser and its desired position is driven to zero as time increases in the presence of the bounded perturbations.)

The sliding variable $\hat{s}$ is chosen as in [73] to have the form

$$\begin{aligned} \hat{s}^s &= \hat{\omega}_{BD}^B + [K_\alpha \star (\hat{q}_{BD}^s (\hat{q}_{BD}^s - \sigma(q_{BD_0}) \hat{1}^s))]^s, \\ \sigma(x) &= \begin{cases} 1, & x \geq 0 \\ -1, & x < 0 \end{cases}, \end{aligned} \quad (6)$$

where q_{BD_0} is the scalar component of the real part of $\hat{q}_{BD}$, and $\hat{s} = s_r + \epsilon s_d$, and thus consequently $\hat{s}^s = s_d + \epsilon s_r$. As in [71], $\hat{s} = s_r + \epsilon s_d = (0, \bar{s}_r) + \epsilon(0, \bar{s}_d)$, where $\bar{s}_r = \bar{\omega}_{BD}^B + \alpha_r \bar{q}_{BD}$ and $\bar{s}_d = \dot{\bar{r}}_{BD}^B + \bar{\omega}_{BD}^B \times \bar{r}_{BD}^B + \frac{1}{2} \alpha_d \bar{r}_{BD}^B$ and the matrices $\alpha_r = \text{diag}(\alpha_{r1}, \alpha_{r2}, \alpha_{r3})$ and $\alpha_d = \text{diag}(\alpha_{d1}, \alpha_{d2}, \alpha_{d3})$, are both symmetric and positive definite.

Three adaptive SMC algorithms are considered and compared in simulations.

1. A *dual-layer adaptive continuous unit vector control* (DAUV) for $\hat{f}_c^B$, that drives the sliding variable $\hat{s}^s$ in Equation (6) to zero as time increases in the presence of differentiable bounded perturbations with unknown bounds, is proposed as

$$\begin{aligned}\hat{f}_c^B &= -(\hat{k} + \eta) \int M^B \star \frac{\hat{S}}{\|\hat{S}\|} dt \\ \hat{s} &= \hat{s}^s + C \star \hat{s}^s, \hat{k} = -\hat{\rho} \text{sign}(\hat{\delta}), \hat{\rho} = r_\alpha + \hat{r} \\ \hat{r} &= \begin{cases} \gamma \|\hat{\delta}\|, & \text{if } \|\hat{\delta}\| > \delta_\alpha \\ 0, & \text{otherwise} \end{cases} \\ \hat{\delta} &= \hat{k} - \frac{1}{\alpha} \left\| \hat{f}_{v,eq}^{B1} \right\| - \zeta, \hat{f}_{v,eq}^{B1} \\ &= \frac{1}{\tau} \left(-(\hat{k} + \eta) \text{sign}(\hat{S})_i - \hat{f}_{v,eq}^{B1} \right) \\ \hat{f}_v^{B1} &= -(\hat{k} + \eta) \int \frac{\hat{S}}{\|\hat{S}\|} dt, \frac{1}{4} \zeta^2 > \delta_\alpha^2 + \frac{1}{\gamma} \left(\frac{q f_1}{\alpha} \right)^2, \quad (7)\end{aligned}$$

where $C = \text{diag}(c_i)$, $c_i > 0$, $i = 1, 2, \dots, 8$, $0 < \alpha < 1$, $q > 1$, γ , r_α , δ_α , η , ζ , $\tau > 0$ are custom chosen constants.

2. A *dual-layer Adaptive Super-Twisting Control* (DAST) [74] for $\hat{f}_c^B$ that drives the sliding variable $\hat{s}^s$ in Equation (6) into a second-order sliding mode (2-SM) in finite time in the presence of differentiable bounded perturbations, for which the bounds are unknown, is given by

$$\begin{aligned}\hat{f}_c^B &= M^B \hat{f}_c^{B1}, \hat{f}_{ci}^{B1} = -\tilde{\lambda}_i \|\hat{s}^s\|^{\frac{1}{2}} \text{sign}(\hat{s}^s)_i \\ &+ \hat{z}_i, \hat{z}_i = -\hat{k}_i \text{sign}(\hat{s}^s)_i, \\ \hat{f}_c^{B1} &= [f_{c1}^{B1}, f_{c2}^{B1}, \dots, f_{c8}^{B1}]^T, \\ \hat{k}_i &= -\hat{\rho}_i \text{sign}(\hat{\delta})_i, \quad \hat{k}_i(0) > 0, \hat{\rho}_i = r_{\alpha i} + \hat{r}_i, \\ \hat{r}_i &= \begin{cases} \gamma_i \|\hat{\delta}_i\|, & \text{if } \|\hat{\delta}_i\| > \delta_{\alpha i} \\ 0, & \text{otherwise} \end{cases} \\ \hat{\delta}_i &= \hat{k}_i - \frac{1}{\alpha_i} \|\hat{w}_{eq}^{B1}\| - \zeta_i, \quad \hat{w}_{eq}^{B1} = \hat{z}_i, \frac{1}{4} \zeta_i^2 > \delta_{\alpha i}^2 \\ &+ \frac{1}{\gamma_i} \left(\hat{q}_i \frac{f_{1i}}{\alpha_i} \right)^2, \zeta_i > 0, \quad (8)\end{aligned}$$

where $\tilde{\lambda}_i > 0$ is sufficiently large, $0 < \alpha_i < 1$, $\hat{q}_i > 1$, and $r_{\alpha i}$, ζ_i , γ_i , $\delta_{\alpha i}$, > 0 are user-definable parameters for all $i = 1, 2, \dots, 8$.

3. A *single-layer adaptive super-twisting control* (SAST) [75] for $\hat{f}_c^B$, which is designed channel by channel as $\hat{f}_{ci}^{B1}$ that drives $\hat{s}^s$ and $\hat{s}^s$ to a real 2-SM in finite time, in which $\|\hat{s}^s\| \leq \eta_{1i}$, $\|\hat{s}_i^s\| \leq \eta_{2i}$, where η_{1i} and η_{2i} for all $i = 1, 2, \dots, 8$ are designer-chosen small constants. This is achieved in the presence of differentiable bounded perturbations using

$$\begin{aligned}\hat{f}_c^B &= M^B \hat{f}_c^B, \hat{f}_{ci}^{B1} = \hat{\alpha}_i \|\hat{s}^s\|^{\frac{1}{2}} \text{sign}(\hat{s}^s)_i + \hat{v}_i, \hat{v}_i = \frac{\hat{\beta}_i}{2} \text{sign}(\hat{s}^s)_i, \\ \hat{f}_{ci}^{B1} &= [f_{c1}^{B1}, f_{c2}^{B1}, \dots, f_{c8}^{B1}]^T,\end{aligned}$$

$$\hat{\alpha}_i = \begin{cases} \omega_0 \sqrt{\frac{\gamma_0}{2}} \text{sign}(\|\hat{s}^s\| - \mu_0)_i, & \hat{\alpha}_i > \hat{\alpha}_{mi} \\ \eta_0, & \hat{\alpha}_i \leq \hat{\alpha}_{mi} \end{cases} \quad \hat{\beta}_i = 2v_i \hat{\alpha}_i, \quad (9)$$

where v_i , ω_0 , γ_0 , η_0 , μ_0 , $\hat{\alpha}_{mi}$ are arbitrary small positive constants that set up the adaptation zone for all $i = 1, 2, \dots, 8$.

The perturbed SF formation, modeled in Equation (5) was simulated under closed-loop control with all the three adaptive SMCs given in Equations (7–9), together with an adaptive classical and 2-SMC [74, 75], and a PD plus controller [73]. Details of the simulation parameters and the results of the simulations may be found in [45]. All five algorithms were executed with a fixed-step of 1×10^{-4} sec using the Euler integration algorithm (i.e., the ode1 solver).

A summary of the results of the simulations is presented in numerical form in Table 1.

The performance characteristics of the novel adaptive-gain SMC algorithms (DAUV, DAST, and SAST) all demonstrate gain non-overestimation. The previously available methods do not surpass the proposed algorithms. Comparing between the newly proposed controllers (7–9), it appears that the SAST has the highest overshoot, while the DAUV and DAST converge with minimal overshoot with respect to the setpoint. The SMC for the perturbed SF presented in [45–48] and discussed above, illustrates the efficacy of the SMCs in controlling the perturbed SFs. Other results of using SMC for controlling both SFs and individual satellites are discussed next.

For perturbed SFs, *continuous HOSM* control is used in [49] in concert with consecutive pulse-width modulation (PWM) of a continuous controller to stabilize the duty cycle of the control thrusters in the presence of bounded disturbances while minimizing fuel consumption.

The enhanced SMC algorithm, specifically *Adaptive SMC*, was applied in [50] for perturbed SF flight based on the nonlinear dynamics of the motion of the follower satellite relative to the leader. The adaptive SMC for perturbed SF flight, which was studied in [51], was based on using neural networks for compensating the SF perturbations. The distributed attitude synchronization control problem was addressed in [52] for SF flight control, in the presence of external disturbances and actuator misalignment, using an adaptive terminal SMC technique. The output tracking control problem of the leader/follower perturbed SF was addressed in [53] using an adaptive terminal SMC law, based on which a modified controller is further developed to solve the dual-equilibrium problem caused by the dual quaternion representation. In [57], attitude synchronization and tracking problems in the presence of the model uncertainties and external disturbances were addressed via a decentralized adaptive SMC. The proposed adaptive SMC was obtained using an undirected inter-spacecraft communication topology and analyzed based on algebraic graph theory.

In [54, 55], *nonlinear* effects are taken into account via Hill's relative motion equations. These are used to model the follower's

TABLE 1 | Control algorithm comparison metrics.

Control Algorithm	Rotational settling Time, sec	Translational settling Time, sec	Peak torque, $N\,M$	Peak force, N	Orientation Integral error	Position integral Error, m	Energy N^2m^2s	Computational Time, sec
PD Plus	66.57	55.15	3.71	5.97	0.75	0.12	445.1	114.2
Adaptive SMC	68.26	66.62	1.88	6.51	0.96	0.15	280.4	51.6
DAUV	17.75	52.23	1.91	1.78	0.88	0.05	130.1	21.7
DAST	11.69	67.23	2.71	4.37	0.77	0.04	244.5	19.1
SAST	36.30	56.71	2.24	3.73	0.83	0.10	265.1	15.1

motion relative to the leader in [54]. A tracking SMC law, which uses propulsive thrusters, is derived to force an SF to follow the natural orbit. In [55], two distributed SMC laws are employed to follow a leader in a desired formation, while compensating unknown nonlinear coupling terms.

The advanced SMC algorithm, specifically a second-order *terminal SMC* (2-SMC), is studied for SF obstacle avoidance in [56] using an artificial potential function. A satellite simulator is used for semi-physical simulation to verify the practical effectiveness of the proposed terminal 2-SMC.

A *triangle tethered SF* system is controlled by fixed-time SMC in [58]. A triangular tethered-satellite formation is formed from three satellites acting as the vertices of a triangle, which are coupled by three elastic tethers. The controller maintains consistency between the satellite formation and between the satellites and tethers. The scheme ensures the settling time is independent of the initial conditions. Coordination terms added to SMC are designed to improve the accuracy of the formation control.

Recently emerged *Fractional-order SMC* is employed in [59] to address the coordinated attitude control problem for SF flying in the presence of disturbances, inertia uncertainty, partial loss of actuators, and communication delays. Compared to traditional SMC methods, the fractional-order SMC algorithm achieve a faster convergence rate.

The efficiency and robustness of SMC methods for perturbed *individual* satellites were investigated in [60–70]. Specifically, in [60], attitude control of an individual satellite using an SMC algorithm was implemented and evaluated on a testbed. The dynamics of the system are derived based on cold-gas thrusters as the actuators, combined with the dynamics of the attitude of the satellite. The control system was evaluated based on both simulation and experimental results. In [61], a decoupled SMC and sliding mode observer (SMO) are proposed for controlling satellite attitude maneuvers in terms of quaternions. The SMO is used to avoid quaternion error differentiation (affected by noise) and to eliminate the requirement for angular velocity measurement.

A *finite-time* continuous SMC is proposed and designed for the attitude control of an earth-pointing magneto-Coulomb satellite in the presence of perturbations, actuated using Coulomb shells and situated in a circular low earth orbit in the presence of the destabilizing gravity gradient torque [62]. A new nonlinear

sliding manifold is proposed in terms of quaternion errors and angular velocity.

The four papers [60, 63–65] investigate the effectiveness of *adaptive* SMC in controlling (perturbed) satellites. A novel *adaptive* continuous fast fixed-time SMC law is proposed in [63] for attitude control of a satellite in the presence of inertia uncertainty and unknown bounded external disturbance torques. The proposed control law mitigates chattering and provides singularity-free satellite attitude control. An *adaptive SMC* law based on dual quaternions for six-degree-of-freedom proximity operations between a chaser and a target spacecraft is designed and investigated in [60]. Dual quaternions are used to model the relative motions during proximity operations. The chattering effect is alleviated by replacing the sign function with a sigmoid. Continuous terminal *adaptive fast SMC* is designed in [52] to control a flexible satellite's position and attitude and to suppress the vibrations of the flexible appendages in the presence of bounded perturbations. An *adaptive* modified SMC is designed in [65] for attitude tracking of a nano-satellite with three magnetorquers and one reaction wheel. A finite-time convergent sliding variable is employed, and the adaptation algorithm helps mitigate chattering.

Sliding mode control laws are designed for polar-orbiting, nadir-pointed satellites in [66]. The proposed SMCs are implemented using only magnetorquers. It is demonstrated that the satellite dynamics with magnetic control are described by a periodically time-varying system, which is stabilized by SMC with a specially designed time-varying sliding variable.

An *event-triggered SMC* for attitude control is designed and studied in [67] for a rigid satellite, to facilitate a track-to-rest maneuver in the presence of bounded external disturbances. In the proposed algorithms, the discretized attitude and control torque signals are fed to the actuator only when the measurement errors surpass a predefined threshold.

A *fractional-order SMC* with enhanced performance is designed for a small satellite with reaction wheels in [68]. The chattering was alleviated by using the sigmoid function approximation. A *fractional neural integral sliding-mode* controller based on the Caputo-Fabrizio derivative and Riemann–Liouville integral for a robot manipulator mounted on a perturbed free-floating satellite is proposed in [69]. A swarm intelligence algorithm is used for tuning the controllers.

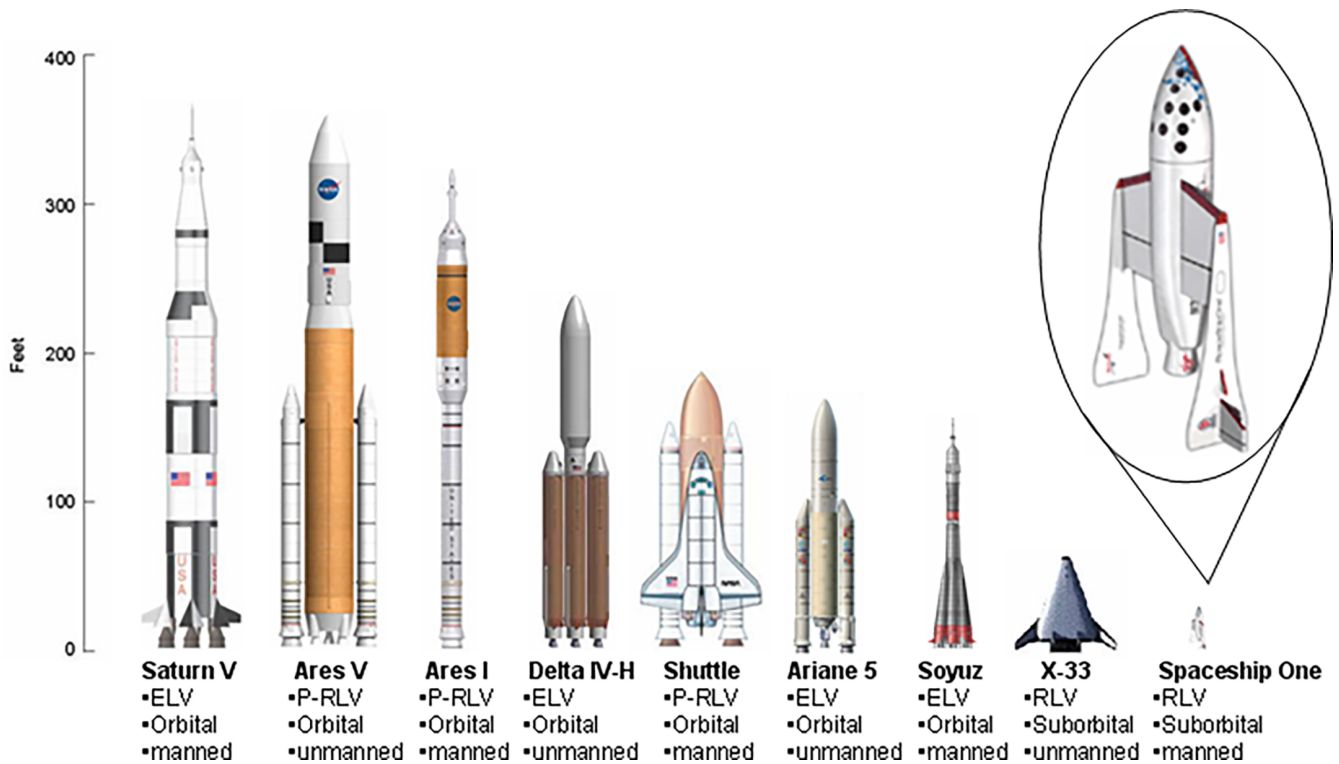


FIGURE 3 | Launch vehicle examples [77, 78].

The design of a finite-time *HOSM controller* for attitude control of a three-axis satellite using solar-radiation pressure is presented in [70]. The satellite, equipped with four solar plates, is assumed to be in an elliptic orbit. The attitude error, as well as its first and second derivatives, converge to the origin in a finite time. A satellite sun-safe controller based on a super-twisting disturbance observer with interval analysis was studied in [76].

Simulation results presented in all the reviewed papers show precise attitude tracking of satellite and satellite formations, when controlled by SMC, SMC-based SMDO, and HOSMC, despite uncertainties in the model and bounded external disturbances. Furthermore, it should be noted that the preceding review shows the widespread use of sliding mode control algorithms for effective robust control of perturbed satellite formations and individual satellites.

4 | Reusable Launch Vehicle Control

Vehicles, with the primary purpose of carrying a payload from the surface of the Earth into outer space, are known as *Launch Vehicles* (LV) [77]. Outer space is defined as everything beyond the Kármán line, that is, 100 kilometers (≈ 62 miles) above the Earth's surface. LVs can support flight at altitudes above the Kármán line since they use thrust instead of aerodynamic lift to overcome the force of gravity.

Reusable Launch Vehicles (RLVs), or reusable components of Partially Reusable Launch Vehicles (P-RLVs), are designed to perform many missions within their life cycle. Orbital LVs can carry a

payload into an orbit about the Earth. Several examples of launch vehicles [77, 78] are shown in Figure 3.

Currently, the most common control algorithm used in their deployment is Proportional Integral Derivative (PID) control with gain scheduling. PID control was used on the Saturn V and the Space Shuttle, and is being developed in the Ares rockets. Although many theoretical control developments have taken place over the last 50 years, PID control with gain scheduling remains dominant due to its reliability.

Technically, the SMC algorithms are robust techniques that can obviate the need for gain scheduling in the RLV controller. It is an attractive design methodology for RLV because of its inherent insensitivity and robustness to bounded plant uncertainties and external disturbances. Furthermore, its robustness can accommodate an RLV's different trajectories, aerodynamic surface and engine failures without the need for gain scheduling. HOSMCs are even more powerful than classical SMCs, and potentially have the capability to handle RLV models with arbitrary relative degree, which providing increased control precision. The application of SMC/HOSMC for the RLV control can be found in many papers, including [77–91].

Under the NASA Space Launch Initiative, Lockheed Martin designed the RLV X-33 (Figures 4 and 5), which was an unmanned, sub-scale technology demonstrator for the Venture Star RLV, which was intended to be a single-stage-to-orbit reusable launch system.

At the beginning of the 2000s, NASA ran the Reusable Launch Vehicle Program that included work on the guidance and

control of the RLV X-33. As part of this program, attitude control approaches based on SMC and SMC driven by sliding mode disturbance observers (SMDO) were studied [78–81]. The equations of motion of the RLV in the body frame with respect to the North-East-Down (NED) frame are used to describe RLV kinematics and dynamics. The kinematics are described in terms of Euler angles as follows [78–81]:

$$\dot{\gamma} = \mathbf{R}(\gamma)\omega \quad (10)$$

where the rotational matrix $\mathbf{R}(\gamma)$ is given by

$$\mathbf{R}(\gamma) = \begin{bmatrix} 1 & \tan \theta \sin \varphi & \tan \theta \cos \varphi \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \frac{\sin \varphi}{\cos \theta} & \frac{\cos \varphi}{\cos \theta} \end{bmatrix}, \quad \gamma = [\varphi \ \theta \ \psi]^T \quad (11)$$

for the ascent and

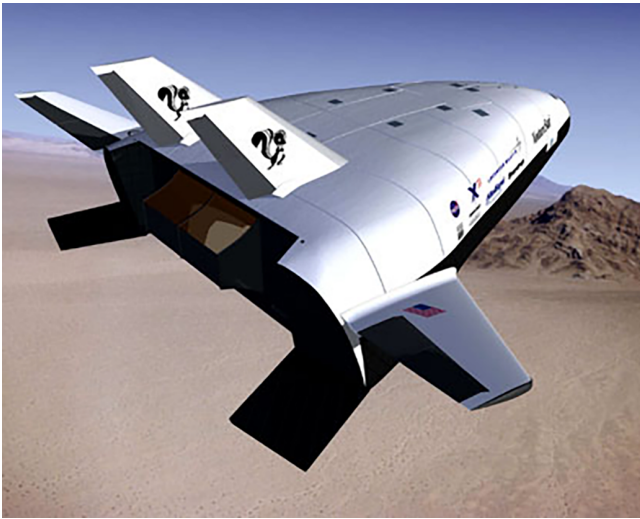


FIGURE 4 | The X-33 RLV [78].

$$\mathbf{R}(\gamma) = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & -\cos \alpha \end{bmatrix}, \quad \gamma = [\varphi_b \ \alpha \ \beta]^T, \quad (12)$$

for entry, where φ, θ , and ψ are the roll, pitch, and yaw angles, respectively, φ_b, α , and β are bank angle, angle of attack, and sideslip angle, respectively. The perturbed dynamics of the RLV can be presented as

$$\dot{\omega} = \mathbf{J}_0^{-1} \boldsymbol{\Omega} \mathbf{J}_0 \omega - \mathbf{J}_0^{-1} \dot{\mathbf{J}}_0 \omega + \mathbf{J}_0^{-1} \mathbf{T} + \mathbf{F}(\cdot), \quad (13)$$

where

$$\boldsymbol{\Omega} = \begin{bmatrix} 0 & r & -q \\ -r & 0 & p \\ q & -p & 0 \end{bmatrix}, \quad \mathbf{J}_0 = \begin{bmatrix} J_{xx} & -J_{xy} & -J_{xz} \\ -J_{xy} & J_{yy} & -J_{yz} \\ -J_{xz} & -J_{yz} & J_{zz} \end{bmatrix}$$

is a known matrix of inertia, and $\mathbf{F}(\cdot) \in \mathbb{R}^3$ is a cumulative norm bounded perturbation vector that includes external disturbances and RLV parameter uncertainties. Here, it is assumed that $\dot{\mathbf{J}}_0 \approx \mathbf{0}$ and the term $\mathbf{J}_0^{-1} \dot{\mathbf{J}}_0 \omega$ can be neglected.

The control problem for the RLV is to determine the control torque vector $\mathbf{T} \in \mathbb{R}^3$ in Equation (13) such that the commanded orientation/attitude profile $\gamma_c(t)$ is robustly asymptotically followed: That is, $\gamma_c(t) - \gamma(t) \rightarrow 0$ as time increases in the presence of the bounded disturbance torque, RLV inertia variations, aerodynamic surface, and engine failures that are contained in $\mathbf{F}(\cdot)$.

The output tracking error is introduced as

$$\mathbf{e}_\gamma = \gamma_c - \gamma, \quad (14)$$

where γ_c is a command attitude angle profile.

A double-loop SMC driven by an SMDO was designed in two steps. Firstly, the RLV body rate vector ω is considered as a *virtual control* in Equations (10) and (13), and is designed as

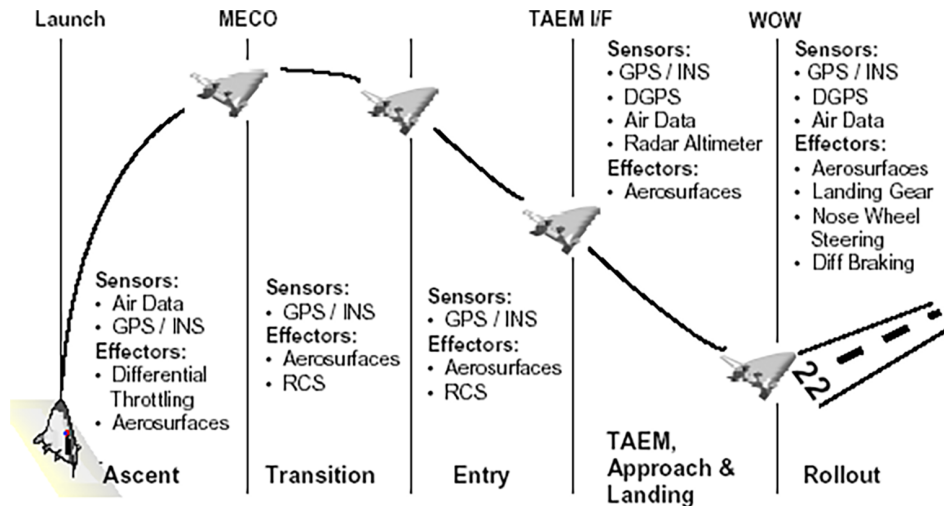


FIGURE 5 | RLV trajectory [78].

$$\omega_c = \mathbf{R}^{-1}(\gamma) [\dot{\gamma}_c + \lambda_1 \mathbf{e}_\gamma], \quad (15)$$

where $\lambda_1 = \text{diag}\{\lambda_{11}, \lambda_{12}, \lambda_{13}\}$, $\lambda_{1i} > 0 \quad \forall i = 1, 2, 3$. Exploiting (10) and (13), the virtual control (15) makes the compensated $\mathbf{e}_\gamma$ dynamics evolve according to

$$\dot{\mathbf{e}}_\gamma = -\lambda_1 \mathbf{e}_\gamma, \quad (16)$$

and therefore $\mathbf{e}_\gamma \rightarrow \mathbf{0}$ as time increases. Note that designing the virtual control (15) does not require the sliding mode format, since Equation (10) is not subject to uncertainties. In contrast, the inner loop dynamics in Equation (13) are perturbed and require a robust controller. The ω dynamics in Equation (13) can be written as

$$\dot{\omega} = \Phi(\cdot) + \mathbf{u}_1, \quad (17)$$

where $\mathbf{u}_1 = \mathbf{J}_0^{-1} \mathbf{T}$ and $\Phi(\cdot) = \mathbf{J}_0^{-1} \Omega \mathbf{J}_0 \omega + \mathbf{F}(\cdot)$ is assumed to be norm-bounded and satisfies $\|\Phi(\cdot)\| \leq L_\Phi$ where $L_\Phi > 0$. In a reasonable flight domain, this is justifiable.

The SMDO for $\Phi(\cdot)$ in Equation (18) is designed as

$$\dot{\mathbf{z}} = \mathbf{u}_1 + \mathbf{u}_2, \quad (18)$$

where $\mathbf{u}_2$ is an injection term. Define the estimation error as $\hat{\mathbf{e}}_\omega = \omega - \mathbf{z}$ then its dynamics evolve according to

$$\dot{\hat{\mathbf{e}}}_\omega = \Phi(\cdot) - \mathbf{u}_2. \quad (19)$$

The injection term is designed in a unit-vector format as

$$\mathbf{u}_2 = \rho \frac{\hat{\mathbf{e}}_\omega}{\|\hat{\mathbf{e}}_\omega\|}, \quad \rho > L_\Phi \quad (20)$$

then $\hat{\mathbf{e}}_\omega \rightarrow \mathbf{0}$ in finite time, and in the sliding mode $\Phi(\cdot) = \mathbf{u}_{2\text{eq}}$. To obtain $\mathbf{u}_{2\text{eq}}$ the entries of $\mathbf{u}_2$ are low-pass-filtered: So, for example, $(\mathbf{u}_{2\text{eq}})_i$, can be approximately estimated as

$$\tau \frac{d(\hat{\mathbf{u}}_{2\text{eq}})_i}{dt} + (\hat{\mathbf{u}}_{2\text{eq}})_i = (\mathbf{u}_2)_i, \quad (21)$$

and $\hat{\Phi}(\cdot) \approx \hat{\mathbf{u}}_{2\text{eq}}$. The control torque $\mathbf{T}$ is designed as follows. Define the output tracking error

$$\mathbf{e}_\omega = \omega_c - \omega, \quad (22)$$

where the sufficiently smooth body rate command profile ω_c is given by (15). The $\mathbf{e}_\omega$ (21) dynamics are given by

$$\dot{\mathbf{e}}_\omega = \dot{\omega}_c - \Phi(\cdot) - \mathbf{u}_1. \quad (23)$$

Next, in the inner (dynamic) loop, the SMC is driven by the SMDO control torque $\mathbf{T}$ is designed for the perturbed RLV dynamics in Equation (13) as

$$\mathbf{T} = \mathbf{J}_0 \left[\dot{\omega}_c - \hat{\Phi}(\cdot) + \lambda_2 \mathbf{e}_\omega \right], \quad (24)$$

where $\lambda_2 = \text{diag}\{\lambda_{21}, \lambda_{22}, \lambda_{23}\}$, $\lambda_{2i} > 0$, $\lambda_{2i} \gg \lambda_{1i} \quad \forall i = 1, 2, 3$; and $\hat{\Phi}(\cdot)$ is the estimate of $\Phi(\cdot)$. Assuming $\hat{\Phi}(\cdot) = \hat{\mathbf{u}}_{2\text{eq}}$

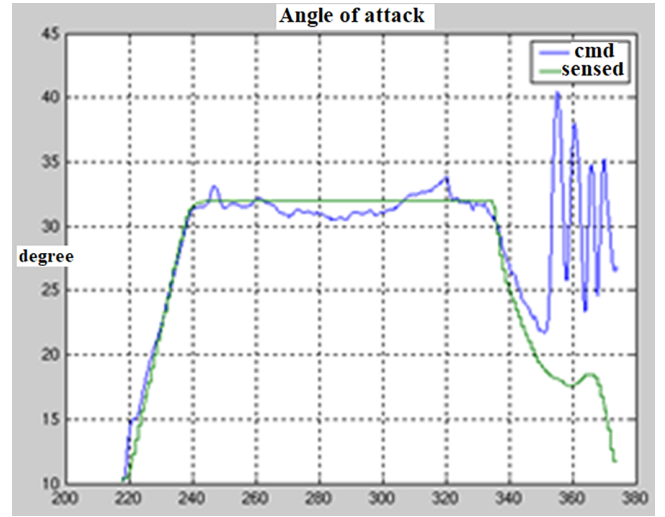


FIGURE 6 | PID Angle of attack control in perturbed mode [78].

exactly, the compensated body rate ω tracking error dynamics are derived as

$$\dot{\mathbf{e}}_\omega = -\lambda_2 \mathbf{e}_\omega, \quad (25)$$

which yields $\mathbf{e}_\omega \rightarrow \mathbf{0}$ as time increases. The control torque $\mathbf{T}$ in Equation (24) generates the actuator command vector δ_c according to $\delta_c = \mathbf{D}^+(\cdot) \mathbf{T}$, where the torque allocation matrix $\mathbf{D}^+(\cdot) = \mathbf{D}(\cdot)^T [\mathbf{D}(\cdot) \mathbf{D}(\cdot)^T]^{-1}$ where $\mathbf{D}(\cdot)$ is the nominal sensitivity matrix.

The results of the simulations of the SMC/SMDO and comparisons with the baseline PID controller in the nominally perturbed case (nominal mode) can be found in [78–81], where one can see a superiority of the SMC/SMDO over the baseline PID controller in terms of the accuracy of the position/attitude tracking.

Next, a specific external perturbation was added and tested via simulations, whereby the right flap fails and remains stuck at +2 deg at 300 s into flight for 20 s. The results of the simulations of the HOSMC and a comparison with the baseline PID controller are presented in Figures 6–9 [78–81]. One can observe that the baseline PID controller fails the test (see Figures 6–9), whereas the SMC/SMDO retains excellent attitude angles tracking performance. Specifically once a special perturbation was applied at 300 s into the flight (1) the PID controller failed to track the angle of attack and the bank angle commands (see Figures 6 and 8); (2) the SMC/SMDO continued to accurately track the angle of attack and bank angle commands (Figures 7 and 9).

Based on the presented analysis, the results of the research on the SMC design for RLV X-33 in launch and entry modes, presented in [85, 86], demonstrated excellent robust performance.

The use of advanced SMC driven by a *second-order* (super-twisting) SMC sliding mode observer (SMC/2-SMDO) for RLV was studied in [78, 79]. The multiple-loop flight controller design is based on disturbance observers. The accuracy of attitude angle tracking was improved dramatically in the presence of the perturbations. A *gain-adaptation* algorithm is included in the disturbance observer design so that it provides the smallest gain needed for the existence of a sliding mode in the observer.

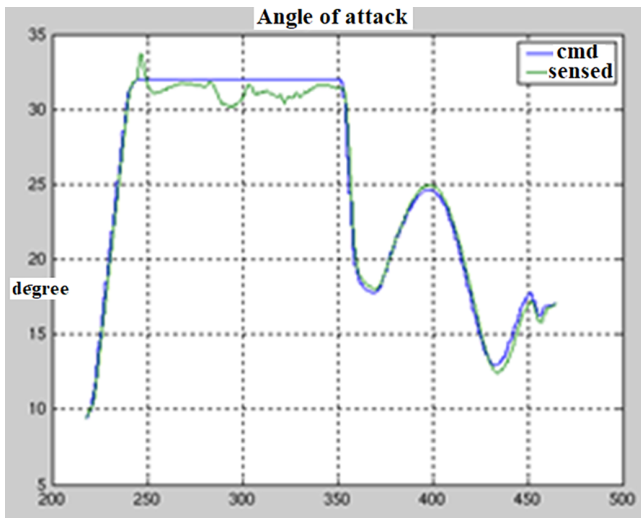


FIGURE 7 | SMC/SMDO Bank angle control (perturbed mode) [78].

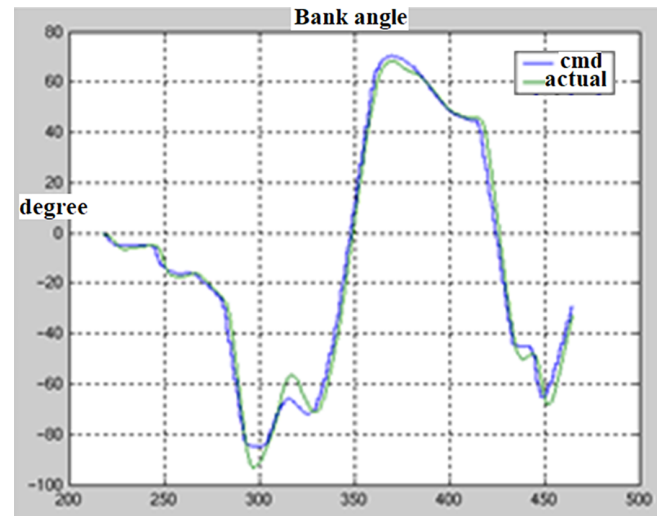


FIGURE 9 | SMC/SMDO Bank angle control (perturbed mode) [78].

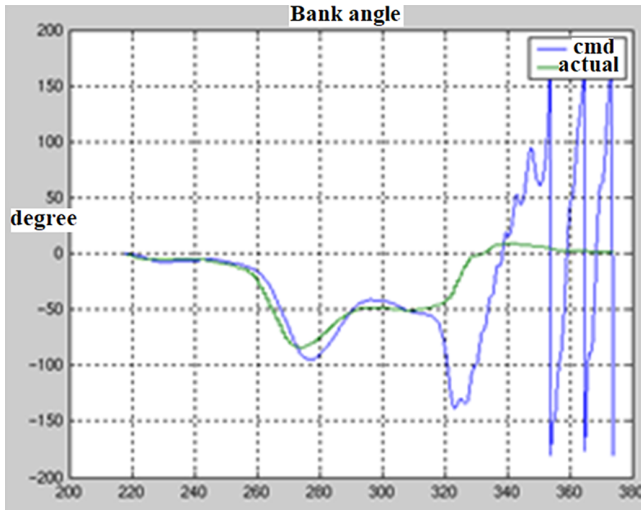


FIGURE 8 | PID Bank angle control in perturbed mode [78].

A *HOSMC* single loop RLV control was investigated in [78, 81]. High-accuracy attitude angle tracking in the presence of bounded disturbances was demonstrated. The effects of the perturbations generated by the flexible and sloshing modes, which were detected in the LV Ares 1, were studied in [83]. These effects were compensated using the SMC/SMDO controller, and robust, high-accuracy attitude angle tracking was achieved.

RLV attitude control in ascent and re-entry flight modes via a multiple time scale *time-varying* SMC driven by a sliding mode observer (TV-SMC-SMO) was studied in [78, 84]. Both the sliding variable and the boundary layer dynamics are designed through proportional-derivative (PD) eigenvalue assignment. This approach is able to cope with control command saturation and integrator windup in the presence of perturbations and control effector failure.

The *air breathing ramjet*-powered RLV is controlled via SMC in [82]. In this work, the problem is aggravated by the large angle of

attack, which yields nonsymmetrical engine operation. A pitch angle SMC is proposed for a conical lifting body RLV.

In [87, 91], *Adaptive* SMC for RLVs was studied. A novel barrier function-based *adaptive* SMC with predefined-time convergence was proposed in [87]. Convergence to a pre-assigned region in a prescribed time was studied, while the adaptive control gain was not overestimated. A Lyapunov method is employed for the stability analysis. In [91], SMC attitude control of the aeroservoelastic RLV in a re-entry mode was designed. The attitude SMC algorithm is implemented in a two-loop format by using the backstepping method. An interval type-2 fuzzy logic *adaptive* control scheme was employed to approximate the nonlinear parts of the model for improving the re-entry attitude tracking performance.

Control of a vertical-landing vertical-takeoff (VTVL) RLV with actuator faults and saturation constraints was considered in [88]. A *fractional-order* predefined-time SMC algorithm with a predefined-time *extended state observer* and a predefined-time anti-saturation compensator was employed for controlling the RLV.

In [89], controlling a six-degree-of-freedom model of an RLV with four types of atmospheric disturbances was considered using SMC techniques. The attitude RLV control was designed using a *nonsingular terminal* SMC.

A *fixed-time non-singular fast terminal* SMC was proposed in [90] for controlling a perturbed VTVL RLV. A fixed-time disturbance observer was developed to estimate uncertainties and disturbances. To prevent the singularity phenomenon, the RLV attitude dynamics were modeled using quaternions. In this work, Model Predictive Control was used to compensate for the delay of the actuator dynamics. SMC-based control for a crewed re-entry vehicle (ACRV) with a small lift/drag ratio $r = 0.3$ was proposed in [92]. The trajectory controller drives the ACRV to a predefined trajectory. The command reference profiles for the attitude controller are generated by the navigation system. The attitude SMC is optimized by using a *genetic algorithm* [92] for optimal disturbance accommodation.

The simulation results presented in all the reviewed papers demonstrate the effectiveness of the proposed SMC, HOSMC, and SMDO-based methods in creating guidance and control laws for RLVs and demonstrate accurate and robust performance. It can be concluded that SMC (including ones with fixed and prescribed convergence time, optimized by genetic algorithms, and with adaptive gains), SMC driven by SMDO, and HOSMC have been widely and effectively used for RLV control in the presence of external disturbances and parametric uncertainties.

5 | Hypersonic Vehicle and Missile Control

A Hypersonic vehicle (HV) is a winged or wingless aircraft with a flight speed exceeding 5 times the speed of sound [93]. It has the advantages of both aircraft and spacecraft, and has significant potential economic and military value. The HV can be of two types: Either a powered or a gliding hypersonic vehicle/missile. Compared to traditional aircraft, the integrated design of HV results in strong coupling between the aerodynamic and propulsion systems. Furthermore, the strong nonlinear characteristics, static instability, together with the fact that the HV is flying at hypersonic velocities within the atmosphere and becomes enveloped in a “plasma sheath”, yielding uncertainties in the aerodynamic coefficients, create huge challenges for the control system design. Also, the external disturbances may result in an actuator failure and engine thermal choking, resulting in system instability.

A large variety of results on HV and hypersonic missile (HM) with Sliding Mode guidance and control laws have been obtained and published during the last 20 years: Including adaptive SMC and HOSMC [94–105], continuous SMC, 2-SMC and HOSMC [106–113], disturbance observer-based SMC [114–116], fractional-order SMC [117], dynamic SMC for nonminimum phase HV [118], discrete time SMC [119], recursive SMC [120], and SMC based on trajectory linearization [121].

Specifically, in [101, 102] control of a perturbed air-breathing HM in the terminal phase of flight is considered. To maximize the ground target penetration, the desired vertical orientation of the HM at the impact should be achieved (Figures 10 and 11) [109].

The Longitudinal dynamics of the rigid body HM is considered in [122] for which an adaptive HOSMC controller was designed.

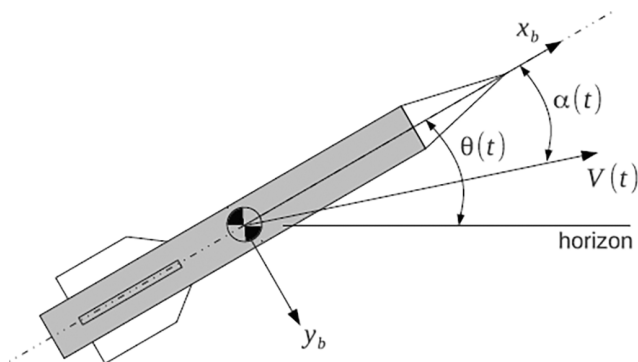


FIGURE 10 | Coordinate frames and variables of HM [109].

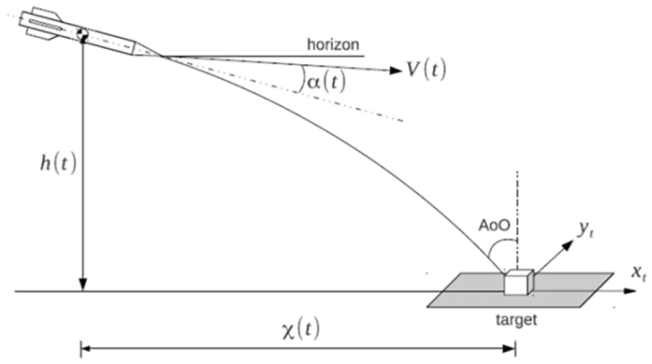


FIGURE 11 | Scenario of HM-target end-game [109].

The mathematical model of the HM longitudinal dynamics considered in [122] is given by

$$\begin{cases} \dot{V} = \frac{T \cos(\alpha) - D}{m} - g \sin(\theta - \alpha) \\ \dot{h} = V \sin(\theta - \alpha) \\ \dot{\alpha} = -\frac{L + T \sin(\alpha)}{mV} + q + \frac{g}{V} \cos(\theta - \alpha) \\ \dot{\theta} = q \\ \dot{q} = \frac{M}{I_{yy}} \\ \dot{\chi} = V \cos(\theta - \alpha) \end{cases}, \quad (26)$$

where $V(t) \in \mathbb{R}$ denotes the forward velocity; $h(t) \in \mathbb{R}$ is the altitude; $\alpha(t) \in \mathbb{R}$ denotes the angle of attack; $\theta(t), q(t) \in \mathbb{R}$ denote the pitch angle and the pitch rate respectively; $\chi(t) \in \mathbb{R}$ is the downrange distance (see Figure 11) measured with respect to the initial HM position; m and $I_{yy} \in \mathbb{R}$ denote the mass and the moment of inertia about the HM body y -axis, respectively; $g \in \mathbb{R}$ is the acceleration due to gravity; $T, D, L \in \mathbb{R}$ represent the thrust, drag, and lift forces, respectively; and $M \in \mathbb{R}$ is the pitching moment about the body y -axis. The flexible modes associated with aero-thermo-elastic effects are neglected for simplicity. The thrust T , drag D , and lift L forces, as well as the pitching moment M about the body y -axis are nonlinear uncertain functions that are given as

$$L = L(\alpha, \delta_e), D = D(\alpha, \delta_e), M = M(T, \alpha, \delta_e), T = T(\phi, \alpha), \quad (27)$$

where $\delta_e(t) \in \mathbb{R}$ and $\phi \in \mathbb{R}$ are the deflection of the elevator surface, and the dimensionless fuel-to-air ratio ($\phi \in [0, 1.5]$) respectively. The control vector is

$$u = [\phi, \delta_e]^T, \quad (28)$$

while the vectors of controlled outputs and command profiles are

$$y = [h, \theta]^T, \quad y_c(t) = [h_c(t), \theta_c(t)]^T. \quad (29)$$

A HOSM control law is designed in terms of derivatives of the control functions (28) to achieve continuity of the control functions as

$$v = [v_1, v_2]^T = [\dot{\phi}, \dot{\delta}_e]^T. \quad (30)$$

Next, the input-output error dynamics of the system in Equations (26), (29), (30) are derived as

$$\begin{cases} e_{\theta}^{(3)} = \theta_c^{(3)}(t) - \underbrace{(\psi_{\theta}(\theta, \phi, \delta_e, \alpha, q, V) + d_{12}\omega_2 + d_{11}\omega_1)}_{\Psi_{\theta}} - \omega_1 \\ e_h^{(3)} = h_c^{(3)}(t) - \underbrace{(\psi_h(\theta, \phi, \delta_e, \alpha, q, V) + d_{21}\omega_1 + d_{22}\omega_2)}_{\Psi_h} - \omega_2 \end{cases}, \quad (31)$$

where

$$e_{\theta} = \theta_c(t) - \theta(t), e_h = h_c(t) - h(t), \quad (32)$$

and

$$G = \begin{bmatrix} \frac{1}{I_{yy}} \frac{\partial M(\phi, \alpha, \delta_e)}{\partial \phi} & \frac{1}{I_{yy}} \frac{\partial M(\phi, \alpha, \delta_e)}{\partial \delta_e} \\ \frac{1}{m} \frac{\partial T(\phi, \alpha)}{\partial \phi} \sin(\theta) & \frac{1}{m} \left[\frac{\partial L(\alpha, \delta_e)}{\partial \delta_e} \cos(\theta - \alpha) - \frac{\partial D(\alpha, \delta_e)}{\partial \delta_e} \sin(\theta - \alpha) \right] \end{bmatrix}, \quad (33)$$

$$G = G_0 + \Delta G = \underbrace{\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}}_{G_0} + \underbrace{\begin{bmatrix} \Delta g_{11} & \Delta g_{12} \\ \Delta g_{21} & \Delta g_{22} \end{bmatrix}}_{\Delta G}, \quad (34)$$

$$G_0 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}, \quad \Delta G \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}^{-1} = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix},$$

$$\begin{aligned} \psi_{\theta}(\theta, \phi, \delta_e, \alpha, q, V) &= \frac{1}{I_{yy}} \frac{\partial M(\phi, \alpha, \delta_e)}{\partial \alpha} \left[-\frac{L(\alpha, \delta_e) + T(\phi, \alpha) \sin(\alpha)}{mV} \right. \\ &\quad \left. + q + \frac{g}{V} \cos(\theta - \alpha) \right] \\ \psi_h(\theta, \phi, \delta_e, \alpha, q, V) &= \frac{1}{m} \left[\frac{\partial T(\phi, \alpha)}{\partial \alpha} \sin(\theta) - \frac{\partial D(\alpha, \delta_e)}{\partial \alpha} \sin(\theta - \alpha) \right] \\ &\quad \left[-\frac{L(\alpha, \delta_e) + T(\phi, \alpha) \sin(\alpha)}{mV} + q + \frac{g}{V} \right. \\ &\quad \left. \cos(\theta - \alpha) \right] - \frac{1}{m} \left[\frac{L(\alpha, \delta_e) + T(\phi, \alpha) \sin(\alpha)}{mV} \right. \\ &\quad \left. - \frac{g}{V} \cos(\theta - \alpha) \right] \\ &\quad [D(\alpha, \delta_e) \cos(\theta - \alpha) + L(\alpha, \delta_e) \sin(\theta - \alpha)] \\ &\quad + \frac{1}{m} T(\phi, \alpha) q \cos(\theta) + \frac{1}{m} \frac{\partial L(\alpha, \delta_e)}{\partial \alpha} \\ &\quad \left[-\frac{L(\alpha, \delta_e) + T(\phi, \alpha) \sin(\alpha)}{mV} \right. \\ &\quad \left. + q + \frac{g}{V} \cos(\theta - \alpha) \right] \cos(\theta - \alpha), \end{aligned} \quad (35)$$

Assumption. The derivatives of the nonlinear terms $\Psi_{\theta}(\theta, \phi, \delta_e, \alpha, q, V, d_{12}, \omega_2)$, $\Psi_h(\theta, \phi, \delta_e, \alpha, q, V, d_{21}, \omega_1)$, and the terms $d_{11}, d_{12}, d_{21}, d_{22}$ are norm-bounded in a reasonable flight domain, that is,

$$\begin{aligned} |\dot{\Psi}_{\theta}(\theta, \phi, \delta_e, \alpha, q, V, d_{12}, \omega_2)| &\leq C_{\theta}, \\ |\dot{\Psi}_h(\theta, \phi, \delta_e, \alpha, q, V, d_{21}, \omega_1)| &\leq C_h, \\ C_{\theta}, C_h &> 0, \end{aligned} \quad (36)$$

and

$$\begin{aligned} |d_{11}| &\leq \varepsilon_{11}, |d_{12}| \leq \varepsilon_{12}, |d_{21}| \leq \varepsilon_{21}, \\ |d_{22}| &\leq \varepsilon_{22}, \varepsilon_{11}, \varepsilon_{12}, \varepsilon_{21}, \varepsilon_{22} > 0, \varepsilon_{11} + \varepsilon_{12} < 1, \varepsilon_{21} + \varepsilon_{22} < 1. \end{aligned} \quad (37)$$

Inequalities (37) guarantee the dominance of the control functions $\omega_i, i = 1, 2$ in the $i^{th}, i = 1, 2$ channels over the perturbations, including cross-coupling between the channels.

A continuous adaptive finite time convergent HOSMCs, which drives $e_{\theta} = \theta_c(t) - \theta(t)$ and $e_h = h_c(t) - h(t)$ to zero in finite time in the presence of bounded perturbations with unknown bounds are defined for the system in Equation (31) in terms of ω_1, ω_2

$$\begin{aligned} \omega_i &= \gamma_{3i} |e_i^{(2)}|^{\alpha_{3i}} \text{sign}(e_i^{(2)}) + \gamma_{2i} |\dot{e}_i|^{\alpha_{2i}} \text{sign}(\dot{e}_i) \\ &\quad + \gamma_{1i} |e_i|^{\alpha_{1i}} \text{sign}(e_i) + \varpi_i, \end{aligned} \quad (38)$$

where $i = \theta, h$, the coefficients $\gamma_{1i}, \gamma_{2i}, \gamma_{3i} > 0$ are selected so that the polynomials $s^3 + \gamma_{3i}s^2 + \gamma_{2i}s + \gamma_{1i}$ are Hurwitz, the coefficients $\alpha_{1i}, \alpha_{2i}, \alpha_{3i}$ are defined according to

$$\alpha_1 = \frac{\bar{\alpha}}{2(2 - \bar{\alpha}) - 1}, \alpha_2 = \frac{\bar{\alpha}}{2 - \bar{\alpha}}, \alpha_3 = \bar{\alpha}, \quad \bar{\alpha} \in (0, 1), \quad (39)$$

and the terms ϖ_i are given by

$$\begin{aligned} \varpi_i &= \eta_{1i} |s_i|^{1/2} \text{sign}(s_i) + \xi_i, \quad \dot{\xi}_i = k_i(t) \text{sign}(s_i), \\ \dot{s}_i &= \dot{e}_i + z_i, \quad \dot{z}_i = -\omega_i - \varpi_i, \end{aligned} \quad (40)$$

where

$$\begin{aligned} \delta_i(t) &= k_i(t) - \frac{1}{\varepsilon_1} |w_{eqi}(t)| - \varepsilon_0, \dot{k}_i(t) = -(\rho_{0i} + \rho_i(t)) \text{sign}(\delta_i(t)), \rho_0 > 0, \\ \dot{\rho}_i(t) &= \gamma |\delta_i(t)|, \quad \gamma > 0. \end{aligned} \quad (41)$$

Then, the control laws (28) is computed as

$$u = \begin{bmatrix} \phi \\ \delta_e \end{bmatrix} = \begin{bmatrix} \int v_1 dt \\ \int v_2 dt \end{bmatrix}; \quad \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = G_0^{-1} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}. \quad (42)$$

A HOSM differentiator that is used to obtain the $\ddot{e}_i, \dot{e}_i$ in Equation (38) is given by

$$\begin{aligned} \dot{z}_{0i} &= -\lambda_4 L_i^{1/5} |z_{0i} - e_i|^{4/5} \text{sign}(z_{0i} - e_i) + z_{1i} \\ \dot{z}_{1i} &= -\lambda_3 L_i^{1/4} |z_{1i} - \dot{z}_{0i}|^{3/4} \text{sign}(z_{1i} - \dot{z}_{0i}) + z_{2i} \\ \dot{z}_{2i} &= -\lambda_2 L_i^{1/3} |z_{2i} - \dot{z}_{1i}|^{2/3} \text{sign}(z_{2i} - \dot{z}_{1i}) + z_{3i} \\ \dot{z}_{3i} &= -\lambda_1 L_i^{1/2} |z_{3i} - \dot{z}_{2i}|^{1/2} \text{sign}(z_{3i} - \dot{z}_{2i}) + z_{4i} \\ \dot{z}_{4i} &= -\lambda_0 L_i \text{sign}(z_{4i} - \dot{z}_{3i}), \end{aligned} \quad (43)$$

where the λ values are taken as $\lambda_0 = 1.1, \lambda_1 = 1.5, \lambda_3 = 3, \lambda_4 = 8$, and $|e_i^{(3)}| \leq L_i, i = h, \theta$. The differentiator (43) converges in finite time to $z_{0i} = e_i, z_{1i} = \dot{e}_i, z_{2i} = \ddot{e}_i$.

The parameters of the simulated model and the simulation results for the linearized HM model given in [109] can be found in [101].

The adaptive high-frequency switching terms that are hidden behind the integrals in Equation (42) are continuous. Their amplitudes adjust to compensate for the additive and multiplicative disturbances and are in line with the adaptive gains $k_h(t)$ and $k_{\theta}(t)$. Note that the first and second layer adaptive gains are bounded, and the control gains $k_h(t)$ and $k_{\theta}(t)$ are not overestimated as shown via simulations in [109]. Therefore, the

application of the proposed adaptive continuous HOSMC to HM control is feasible and achieves excellent results in the presence of bounded additive and multiplicative disturbances, whose nature, format, and boundaries are unknown.

The adaptive SMC algorithms proved very promising in aerospace applications as we discussed in Sections 2–4. One of the first results in HV control using *adaptive* SMC is presented in [105]. The multi-input-multi-output (MIMO) *adaptive* SMC is designed for a generic unstable system with uncertain parameters. Simulation studies confirm robust, accurate output tracking of the uncertain HV and show reduced amplitude in terms of the control inputs. A fixed time disturbance observer is used in concert with an *adaptive* HOSMC for HM in [100] and demonstrates high accuracy reconstruction of the perturbations.

A dynamic *adaptive* SMC algorithm for HV was developed and proposed in [94]. A mathematical model for the uncertainty of the HV is derived, and its bounds are reconstructed by the adaptive SMO. Artificial potential functions are used for online estimation of control saturation and response rate.

A novel *multivariable adaptive* super-twisting controller was designed and proposed for controlling perturbed/uncertain HVs in [96]. The continuity of the control signals and finite time convergence are guaranteed. A novel adaptive control law was developed by analyzing the connection between the control gains and the boundaries of the perturbations. The closed-loop control system is proved to be globally ultimately stable.

A disturbance observer-based *adaptation HOSMC* algorithm is proposed in [115] for HVs with bounded perturbations, whose bounds are unknown. The disturbance observer with gain adaptation is added to the HOSMC, which compensates for the estimation errors. As a result, unnecessarily large gains for maintaining robustness are avoided, and control chattering is mitigated.

Control of a Hypersonic Glider (HG) was studied in [97]. A novel autopilot based on a continuous adaptive-gain HOSMC algorithm performing in conjunction with impulsive reaction control thrusters was proposed. Impulsive control enables almost instantaneous compensation for the initial conditions. Likewise, almost instantaneous convergence is provided by the impulsive augmentation of the continuous HOSMC. The risk of control loss caused by sideslip angle departures, due to poor transversal stability and small lateral control, authority is mitigated.

The robust flight control problem for the longitudinal dynamics of a generic HV in the presence of mismatched disturbances was considered in [98]. An adaptive SMC algorithm with a backstepping-based nonlinear disturbance observer was proposed and studied.

In [99], an *adaptive*, SMC based on an adaptive SMDO was proposed for controlling an uncertain hypersonic entry vehicle in the presence of atmospheric density variations, inertia, and aerodynamic uncertainties.

An *adaptive* continuous nonsingular terminal SMC was proposed in [95] for hypersonic glide vehicle control. The perturbations

were reconstructed by using a super-twisting SMDO. The proposed adaptive law guarantees finite-time convergence to a real 2-SMC without gain over-estimation.

Time-varying sliding mode based fixed-time missile guidance and attitude control [108], quasi-continuous HOSMC and SMDO in [110], adaptive HOSMC in [103, 104], and an improved continuous SMC algorithm in [106] have all been designed and studied for *flexible* perturbed HV control.

Recursive *terminal* SMC augmented by SMDO, and *terminal* SMCs in concert with SMDOs were proposed for HV control in [111, 120], and [112] respectively. Furthermore, in [112] the backstepping method was employed in the altitude control subsystems.

A *nonminimum phase* HV was controlled by SMC-based SMDO in [118] using a dynamic sliding manifold approach for stabilizing the unstable internal dynamics. A 2-SMC controller was used in [107] for controlling a *nonminimum phase* underactuated HV.

An extreme learning machine (ELM) neural network *disturbance observer* (NNDO)-based on a continuous SMC algorithm was proposed in [114] for precise tracking control under various disturbed scenarios. The ELM-NNDO was designed based on a single-hidden-layer feedforward network. The neural network observer is inherently continuous/smooth, but does not provide exact estimation. Another *SMDO* was designed in [116] to facilitate the control of the longitudinal dynamics of a generic air-breathing HV in the presence of external bounded and model uncertainties. Integral SMC based on a *SMDO* is used in [113] for HV longitudinal control.

A novel hybrid control that combined an extended disturbance observer and a continuous SMC with *trajectory linearization control* was proposed in [121] for addressing the perturbed HV attitude tracking problem.

To follow velocity and acceleration commands for a *hypersonic cruise missile*, a multidimensional *discrete* variable structure control was proposed in [119]. Global finite-time convergence was achieved after employing an appropriate power-rate reaching law.

To guide a low-speed missile for intercepting HM in the longitudinal plane, a fractional-order SMC-based guidance law was proposed in [117]. To precisely estimate the disturbance, a non-homogeneous disturbance observer is designed. Note that the use of SMDO mitigates chattering by means of the SMC's possible gain reduction. A novel fixed-time sliding-mode time-varying guidance and attitude control algorithm was proposed for a missile interceptor [123]. The fixed-time convergence was proven based on the Lyapunov stability theory.

Comparisons and extensive simulation results are presented in all reviewed papers to demonstrate the effectiveness of the proposed SMC, HOSMC and SMDO-based guidance and control laws for controlling perturbed HV and HM. Once again, it can be concluded that these techniques are effective when designing guidance and control laws for different types of HV and HM in

the presence of bounded external disturbances and parametric uncertainties.

6 | Conclusions

A large variety of sliding mode control laws, including conventional and continuous SMC, as well as SMC based on SMDOs, adaptive SMC, and continuous HOSMC, which have been considered for aerospace applications, including control of damaged aircraft, satellite and formation satellite control, reusable launch vehicle control, and the guidance and control of hypersonic vehicles and missiles, have been analyzed in this survey. It was noted that second-order and higher-order sliding mode control systems (2-SMC/HOSMC) retain all the advantages of SMC, such as robustness to matched bounded perturbations and finite time convergence, but in addition can handle systems of higher relative degree, which is typical for aerospace vehicles, while providing control continuity and higher accuracy output tracking error stabilization. The adaptive SMC/2-SMC/HOSMC-based controllers effectively manage aerospace systems, whose perturbations are bounded with unknown bounds. The advantages of SMC/2-SMC/HOSMC-based techniques, including high accuracy finite time convergence, and insensitivity to bounded perturbations and parametric uncertainties, are confirmed via simulations. Future work could be dedicated to more practical studies and tests of the SMC, 2-SMC, and HOSMC algorithms on unmanned and manned flights.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

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